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- 1) For the convolutional layers, to find the number of parameters:

$\# \text{ of params} = (\# \text{ of input channels} \times \text{kernel height} \times \text{kernel width} \times \# \text{ of output channels}) + \# \text{ of output channels}$. For the fully connected layer, to calculate the number of parameters:

$\# \text{ of input features} \times \# \text{ of output features} + \# \text{ of output features}$

The CNN network has seven convolutional layers and one fully connected layer.

Layer	Calculation	Parameters
conv1_1	$(3 \times 3 \times 3 \times 32) + 32$	896
conv1_2	$(32 \times 3 \times 3 \times 32) + 32$	9248
conv2_1	$(32 \times 3 \times 3 \times 64) + 64$	18496
conv2_2	$(64 \times 3 \times 3 \times 64) + 64$	36928
conv3_1	$(64 \times 3 \times 3 \times 128) + 128$	73856
conv3_2	$(128 \times 3 \times 3 \times 128) + 128$	147584
conv4	$(128 \times 3 \times 3 \times 256) + 256$	295168
fc	$(256 \times 11) + 11$	2827

If we add all the parameters of the layers together, we get:

$$896 + 9248 + 18496 + 36928 + 73856 + 147584 + 295168 + 2827 = 585,003$$

Therefore, the CNN network has 585,003 parameters. Now, we can do the same for the CNN_small network:

Layer	Calculation	Parameters
conv1	$(3 \times 3 \times 3 \times 32) + 32$	896
conv2	$(32 \times 3 \times 3 \times 64) + 64$	18496
fc1	$(4096 \times 512) + 512$	2097664
fc2	$(512 \times 10) + 10$	5130

Again, if we add all this together, we get:

$$896 + 18496 + 2097664 + 5130 = 2122186$$

Therefore, the CNN_small network has 2,122,186 parameters.

- 2) The issue with the original network was that it was too simple and therefore unable to learn the necessary patterns from the images. We improved the CIFAR-10 accuracy by adding more layers to the network, which allowed it to pick up on more complex patterns. Also, we made learning easier and faster by including batch normalization to stabilize the training process and max pooling to shrink the images. Lastly, we included a dropout layer to prevent the network from overfitting or memorizing the training data, which improved the final test accuracy to about 78%.